

Experiment – 4.1

Aim: Demonstration of COMMIT, SAVEPOINT and ROLLBACK in CLASS Table

ID	Name
1	Priyanshi
2	Akansha
3	Dolly
4	Priya
5	Aman
6	Himanshi
7	Sakshi

Code:

```
1 • CREATE DATABASE Priyanshi;
2
3 • USE Priyanshi;
4
5 • CREATE TABLE Student (
6     Student_ID INT PRIMARY KEY,
7     Name VARCHAR(50)
8 );
9
10 • INSERT INTO Student VALUES
11     (1, 'Priyanshi'),
12     (2, 'Akansha'),
13     (3, 'Dolly'),
14     (4, 'Priya'),
15     (5, 'Aman'),
16     (6, 'Himanshi'),
17     (7, 'Sakshi');
18
19 • COMMIT;
```

```

21 • UPDATE Student SET Name='Abhijeet'
22   where Student_ID ='5';
23
24 • SELECT * FROM STUDENT;
25
26 • SAVEPOINT A;
27
28 • INSERT INTO Student VALUES
29   (8,'Cheris');
30
31 • SAVEPOINT B;
32
33 • INSERT INTO Student VALUES
34   (9, 'Ram');
35
36 • SAVEPOINT C;
37
38 • SELECT * FROM STUDENT;
39
40 • ROLLBACK TO B;
41
42 • SELECT * FROM Student;
43
44 • ROLLBACK TO A;

```

Output:

UPDATE class SET name = 'Abhijeet' WHERE id = '5';

ID 5 name changes from Rahul -> Abhijeet

1. SELECT 1 Output:

	Student_ID	Name
▶	1	Priyanshi
	2	Akansha
	3	Dolly
	4	Priya
	5	Abhijeet
	6	Himanshi
	7	Sakshi
✱	NULL	NULL

2. ROLLBACK TO B;

SELECT 2 Output:

	Student_ID	Name
▶	1	Priyanshi
	2	Akansha
	3	Dolly
	4	Priya
	5	Abhijeet
	6	Himanshi
	7	Sakshi
	8	Cheris
	9	Ram
*	NULL	NULL

3. ROLLBACK TO A;

Final Table State

	Student_ID	Name
▶	1	Priyanshi
	2	Akansha
	3	Dolly
	4	Priya
	5	Abhijeet
	6	Himanshi
	7	Sakshi
	8	Cheris
	9	Ram
*	NULL	NULL

Experiment – 4.2

Aim: To study and perform Transaction Control Language (TCL) commands such as SAVEPOINT, ROLLBACK, ROLLBACK TO SAVEPOINT, and COMMIT on the given table ACCOUNT_TXN.

Acc_ID	Holder_Name	Balance
101	Aman	5000
102	Bhavna	7000
103	Chirag	6000

Code:

```
1 • CREATE DATABASE Account;
2
3 • USE Account;
4
5 • CREATE TABLE ACCOUNT_TXN (
6     Acc_ID INT PRIMARY KEY,
7     Holder_Name VARCHAR(50),
8     Balance INT
9 );
10
11 • INSERT INTO ACCOUNT_TXN VALUES
12     (101, 'Aman', 5000),
13     (102, 'Bhavna', 7000),
14     (103, 'Chirag', 6000);
15
16 • INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);
17
18 • SAVEPOINT S1;
19
20 • UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101;
21
22 • UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102;
23
24 • SAVEPOINT S2;
25
26 • DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103;
27
28 • INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);
```

```
SAVEPOINT S3;
```

```
UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;
```

```
SELECT * FROM ACCOUNT_TXN;
```

```
ROLLBACK TO S2;
```

```
SELECT * FROM ACCOUNT_TXN;
```

```
INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500);
```

```
SAVEPOINT S4;
```

```
UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);
```

```
SELECT * FROM ACCOUNT_TXN;
```

```
ROLLBACK TO S4;
```

```
SELECT * FROM ACCOUNT_TXN;
```

```
COMMIT;
```

Output:

1. SELECT 1 Output:

	Acc_ID	Holder_Name	Balance
▶	101	Aman	6500
	102	Bhavna Sharma	6000
	104	Diya	4000
	105	Eshan	3000
*	NULL	NULL	NULL

2. Select 2 Output (After ROLLBACK TO S2):

Result Grid			
	Acc_ID	Holder_Name	Balance
▶	101	Aman	6500
	102	Bhawna	6000
	103	Chirag	6000
	104	Diya	4000
*	NULL	NULL	NULL

3. Select 3 output:

Result Grid			 Filter Rows:
	Acc_ID	Holder_Name	Balance
	101	Aman	7000
	102	Bhawna	6000
	103	Chirag	6000
	104	Diya	4500
	106	Farah	4500
	NULL	NULL	NULL

4. Select 4 Output (After ROLLBACK TO S4):

Result Grid			 Filter Rows:	
	Acc_ID	Holder_Name	Balance	
	101	Aman	6500	
	102	Bhawna	6000	
	103	Chirag	6000	
	104	Diya	4000	
	106	Farah	4500	
	NULL	NULL	NULL	

5. Rows Permanently Saved After COMMIT:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna	6000
103	Chirag	6000
104	Diya	4000
105	Farah	4500

1. Difference between ROLLBACK and ROLLBACK TO SAVEPOINT

- ROLLBACK cancels the entire transaction.
- ROLLBACK TO SAVEPOINT cancels changes only up to a specific savepoint.

2. Can we rollback to S3 after ROLLBACK TO S2?

- No.
Because when we rollback to S2, all savepoints created after S2 (like S3) are removed.

3. What happens if COMMIT is executed before creating any savepoint?

- All changes become permanent and cannot be rolled back.